Here's one more pair of results to justify why we will care about meromorphic functions a lot once we define them in a second.

Maximum Modulus Theorem: Suppose M is a connected Riemann surface and $F : M \to \mathbb{C}$ is a holomorphic function. If $|F|$ attains a local maximum on M then F is a constant function.

Proof:

Suppose $|F|$ has a local maximum at $p \in M$. Then let (U, φ) be a complex chart containing p. By shrinking U if necessary, we can also without loss of generality assume $\varphi(U)$ is a connected set. But now $F \circ \varphi^{-1}$ is a holomorphic function on the region $\varphi(U)$ of $\mathbb{C}$ which also satisfies that $|F \circ \varphi^{-1}|$ attains a local maximum at $\varphi(p)$. It follows by the maximum modulus principle from complex analysis (see *page 384*) that $F \circ \varphi^{-1}$ is a constant function on $\varphi(U)$. In turn $F \circ \varphi^{-1} \circ \varphi = F$ is also a constant function on U. And by the identity principle, we can then conclude that F is constant on M. ■

As a side note since I realize that I never showed this, any constant function between Riemann surfaces is automatically holomorphic. To see why, suppose M and N are Riemann surfaces, and $F : M \to N$ satisfies that $F(p) = q \in N$ for all $p \in M$. Then having picked a complex chart (V, ψ) in N containing Q, we have for every complex chart (U, φ) in M that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a constant function from $\varphi(U) \subseteq \mathbb{C}$ into $\mathbb{C}$. So, $\psi \circ F \circ \varphi^{-1}$ is trivially holomorphic.

Corollary: If M is a compact Riemann surface then $O(M, \mathbb{C})$ consists entirely of locally constant functions. In particular, if M is connected then $O(M, \mathbb{C})$ consists only of constant functions.

Proof:

Note that every connected component of M is closed. And since closed subsets of compact sets are compact, we can thus conclude that every connected component of M is compact.

Also note that since M is locally connected, we know that every connected component of M is open. Therefore every connected component of M (equipped with the subspace topology) can sill still be viewed as a Riemann surface where F restricted to each component is holomorphic.

Finally, suppose $F \in O(M, \mathbb{C})$ and let N be any connected component of M. Then F being continuous on N plus N being compact implies that $|F|$ attains a local maximum on N. Yet also note that F being holomorphic on N as a Riemann surface means that the maximum modulus theorem applies. So, we must have that F is constant on N. ■.

Since a lot of the Riemann surfaces we actually care about are compact and connected, we thus can see that $O(M, \mathbb{C})$ is often not the collection of functions we want to study on a Riemann surface. Hence, we will instead often consider $O(M, \mathbb{C}_\infty)$.

To motivate the next bit, observe that if $U \subseteq \mathbb{C}$ is an open set then any meromorphic function $f : U \to \mathbb{C}$ extends to a holomorphic function from U into $\mathbb{C}_\infty$ (in the Riemann surface sense).

If you squint you can see that this is just my comment from *page 467*. After all, $z \mapsto f(z)$ is a holomorphic function from $U - f^{-1}(\{\infty\})$ into $\mathbb{C}$ and $z \mapsto 1/f(z)$ is a holomorphic function from $U - f^{-1}(\{0\})$ into $\mathbb{C}$.

Conversely, suppose $G \subseteq \mathbb{C}$ is a region and $f : G \to \mathbb{C}_\infty$ is a holomorphic function (in the Riemann surface sense) satisfying that there exists $z \in G$ with $f(z) \neq \infty$. Since f is not the constant ∞-valued function, we know by the identity principle that $f^{-1}(\{\infty\})$ can't have any limit points. This also means every $a \in f^{-1}(\{\infty\})$ must be an isolated point of G.

Let $\Omega := G - f^{-1}(\{\infty\})$. Since f is continuous, we can conclude that Ω is an open subset of G. So, we consider the chart $(\Omega, \mathrm{Id}_\Omega)$. Next, we consider the chart $(\mathbb{C}, \mathrm{Id}_\mathbb{C})$ in the standard structure on $\mathbb{C}_\infty$. So, $\mathrm{Id}_\mathbb{C} \circ f \circ \mathrm{Id}_\Omega = f$ is a holomorphic function from Ω into $\mathbb{C}$ (in the complex analytic sense).

But now $f|_\Omega$ has singularities at every $a \in f^{-1}(\{\infty\})$. Also by the continuity of f on G, we know that $|f(z)| \to \infty$ as $z \to a$. Therefore, we can conclude that $f|_\Omega$ has poles at all $a \in G - \Omega$. Or in other words, f is a meromorphic function on G (in the complex analytic sense).

All in all, we can conclude that if $U \subseteq \mathbb{C}$ is an open set, then the set of meromorphic functions on U is precisely equal to $O(U, \mathbb{C}_\infty) - \mathcal{F}$ where $\mathcal{F}$ is the collection of holomorphic functions that are constantly ∞ on any neighborhood. For the time being, I'm going to denote the set of meromorphic functions on U as $\mathcal{M}(U)$.

I realize that $\mathcal{M}$ and M look similar. So, I'll probably start denoting Riemann surfaces by X and Y instead.

6/29/2026

Proposition: Let X be a connected Riemann surface. And suppose $f : X \to \mathbb{C}_\infty$ is a function such that $f \not\equiv \infty$ on X. Then the following are equivalent.

(a) $f : X \to \mathbb{C}_\infty$ is a holomorphic function.

(b) For all complex charts (U, φ) on X we have that $f \circ \varphi^{-1}$ is a meromorphic function on $\varphi(U)$.

(c) For every $p \in X$ there exists a complex chart (U, φ) on X with $p \in U$ and $f \circ \varphi^{-1}$ a meromorphic function on $\varphi(U)$.

$(a \implies b)$
Let (U, φ) be any complex chart on X. Since f is continuous and $f \not\equiv \infty$ on X, we know that $f^{-1}(\{\infty\}) \cap U$ must be a closed set (relative to U) with no limit points in U. In other words, $\Omega := U - f^{-1}(\{\infty\})$ is an open set with $f(\Omega) \subseteq \mathbb{C}_\infty - \{\infty\}$. If we now consider

the charts $(\Omega, \varphi|_\Omega)$ and $(\mathbb{C}, \mathrm{Id})$, we know that $\mathrm{Id} \circ f \circ \varphi^{-1} = f \circ \varphi^{-1}$ is a holomorphic map from $\varphi(\Omega) = \varphi(U) - \varphi(f^{-1}(\{\infty\}))$ into $\mathbb{C}$ since f is holomorphic in the Riemann surface sense. Furthermore, we know every $a \in \varphi(U - \Omega)$ has a neigborhood N where $f \circ \varphi^{-1}(z) \neq \infty$ for any $z \in N - \{a\}$. Plus by the continuity of $f \circ \varphi^{-1}$, we know that $f \circ \varphi^{-1}(z) \to \infty$ as $z \to a$. So, every $a \in \varphi(U - \Omega)$ is a pole. And this finishes showing that $f \circ \varphi^{-1}$ is meromorphic on $\varphi(U)$.

$(b \implies c)$
This is obvious.

$(c \implies a)$
Suppose $p \in X$ and let (U, φ) be a complex chart on X such that $p \in U$ and $f \circ \varphi^{-1}$ a meromorphic function on $\varphi(U)$. If $f(p) \neq \infty$ then by shrinking U and $\varphi(U)$ to a coordinate ball around p, we can assume $f \neq \infty$ anywhere on B. And now $f(U) \subseteq \mathbb{C}$ and $\mathrm{Id} \circ f \circ \varphi^{-1} = f \circ \varphi^{-1}$ is a holomomorphic function on $\varphi(U)$.

Meanwhile, if $f(p) = \infty$ then we know $\varphi(p)$ is a pole of $f \circ \varphi^{-1}$. So, we can shrink U and $\varphi(U)$ to a coordinate ball so that $f \neq 0$ anywhere on U. Next, $f(U) \subseteq \mathbb{C}_\infty - \{0\}$ and $z \mapsto 1/f(\varphi^{-1}(z))$ is a holomorphic function on $\varphi(U)$. ■

Given a Riemann surface X, we say $f : X \to \mathbb{C}_\infty$ is <u>meromorphic</u> if $f \in O(X, \mathbb{C}_\infty)$ and $f \not\equiv \infty$ on any neighborhood in X. We denote the set of meromorphic functions on X as $\mathcal{M}(X)$.

Example: What are the meromorphic functions on $\mathbb{C}_\infty$?

Suppose $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ is meromorphic. Since $\mathbb{C}_\infty$ is compact, we know that f has only finitely many <u>poles</u> (i.e. points in the set $f^{-1}(\{\infty\})$). So, we can let $p_1, \ldots, p_n$ be the poles of f in $\mathbb{C}$ (repeated according to multiplicity). Then $g(z) := f(z)\prod_{i=1}^n (z - p_i)$ is a well-defined entire function on $\mathbb{C}$.

Next, let $g(z) = \sum_{k=0}^\infty c_k z^k$ be the Taylor expansion of g about 0. Since g is an entire function, we know this power series has an infinite radius of convergence. In turn, we also know $g(\frac{1}{z}) = \sum_{k=0}^\infty c_k z^{-k}$ is absolutely convergent for all $z \in \mathbb{C} - \{0\}$. Yet also note $g(\frac{1}{z}) = f(\frac{1}{z})\prod_{i=1}^n (\frac{1}{z} - p_i)$. And since f is meromorphic, we know that $h(z) := f(\frac{1}{z})$ has either a removable singularity or pole at $z = 0$. It follows that $z \mapsto g(\frac{1}{z})$ has a pole or removable singularity at $z = 0$ as well.

Yet now we can conclude there exists $K \in \mathbb{N}$ with $c_k = 0$ for all $k \geq K$. (If not then $z \mapsto g(\frac{1}{z})$ would have an essential singularity.) So, we've proven there exists a polynomial $P(z) \in \mathbb{C}[z]$ with $P(z) = f(z)\prod_{i=1}^n (z - p_i)$. Finally, by rearranging our expression and factoring $P(z)$, we can conclude that there exists $\lambda \in \mathbb{C}$ as well as $a_1, \ldots, a_m \in \mathbb{C}$ such that:

$$f(z) = \lambda \tfrac{(z-a_1)\cdots(z-a_m)}{(z-p_1)\cdots(z-p_n)}$$

So, $\mathcal{M}(\mathbb{C}_\infty) \subseteq \mathbb{C}(z)$ (where $\mathbb{C}(z)$ is the field of rational functions in $\mathbb{C}$). Conversely, it's easy to show every rational function is a holomorphic map from $\mathbb{C}_\infty$ into $\mathbb{C}_\infty$. ■

Here's another interesting observation that the prior example brought to mind.

Lemma: Suppose X and Y are connected Riemann surface with X also compact. Then any nonconstant holomorphic function $F \in O(X, Y)$ must be surjective.

Proof:
Since F is continuous and X is compact, we know that $F(X)$ must be a compact subset of Y. In turn, since Y is Hausdorff we also know that $F(X)$ is a closed subset of Y. Meanwhile, since F is a nonconstant holomorphic function and X is connected, we know F is an open map. In turn, as X is open we must have that $F(X)$ is an open subset of Y. Finally, we trivially know that $F(X) \neq \emptyset$. So, we can conclude from the connectedness of Y that $F(X) = Y$. ■

As a result of this lemma, any nonconstant rational function in $\mathbb{C}(z)$ must define a surjective map from $\mathbb{C}_\infty$ into $\mathbb{C}_\infty$. In particular, this means that every nonconstant meromorphic function on $\mathbb{C}_\infty$ will have both poles and zeros.

6/30/2026

I moved to the room next door today. I was hoping to cover some category theory but all I managed to cover were the following definitions. I was not very productive today. Firstly, here are two definitions I got from *Algebra: Chapter 0* by Paulo Aluffi. (see *citation 8*...)

Suppose $\mathbf{I}$ is category (which we can think of as an indexing category). Next, we consider a functor $F : \mathbf{I} \to \mathbf{C}$. The limit of F (if it exists) is defined to be an object L in $\mathbf{C}$ along with a class of morphisms $\pi_I \in \mathrm{Hom}_{\mathbf{C}}(L, F(I))$ such that:

1. the following diagram commutes for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{End}_{\mathbf{I}}(I, J)$

$$\begin{tikzcd}
& L \arrow[dl, "\pi_I"'] \arrow[dr, "\pi_J"] & \\
F(I) \arrow[rr, "F(f)"'] & & F(J)
\end{tikzcd}$$

2. If $M \in \mathrm{Obj}(\mathbf{C})$ and $\eta_I \in \mathrm{Hom}_{\mathbf{C}}(M, F(I))$ is any other object and collection of morphisms satisfying the prior commuting diagram, then there exists a unique morphism $g : M \to L$ such that:

$$\begin{tikzcd}
& M \arrow[d, dashed, "\exists! g"] \arrow[ddl, bend right, "\eta_I"'] \arrow[ddr, bend left, "\eta_J"] & \\
& L \arrow[dl, "\pi_I"'] \arrow[dr, "\pi_J"] & \\
F(I) \arrow[rr, "F(f)"'] & & F(J)
\end{tikzcd}$$

Remark: It's easy to see any two limits of a functor $F : \mathbf{I} \to \mathbf{C}$ are isomorphic. Thus, we can refer to there being a single limit (denoted as $\varprojlim(F)$). Also, we call this limit a projective limit.

Similarly, the colimit of F (if it exists) is defined to be an object C in **C** along with a class of morphisms $j_I \in \mathrm{Hom}_{\mathbf{C}}(F(I), C)$ such that:

1. the following diagram commutes for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{End}_{\mathbf{I}}(I, J)$

$$\begin{tikzcd} F(I) \arrow[rr, "F(f)"] \arrow[dr, "j_I"'] & & F(J) \arrow[dl, "j_J"] \\ & L & \end{tikzcd}$$

2. If $M \in \mathrm{Obj}(\mathbf{C})$ and $\eta_I \in \mathrm{Hom}_{\mathbf{C}}(M, F(I))$ is any other object and collection of morphisms satisfying the prior commuting diagram, then there exists a unique morphism $g : C \to M$ such that:

$$\begin{tikzcd} F(I) \arrow[rr, "F(f)"] \arrow[dr, "j_I"'] \arrow[ddr, bend right, "\eta_I"'] & & F(J) \arrow[dl, "j_J"] \arrow[ddl, bend left, "\eta_J"] \\ & L \arrow[d, dashed, "\exists! g"] & \\ & M & \end{tikzcd}$$

Remark: It's easy to see any two colimits of a functor $F : \mathbf{I} \to \mathbf{C}$ are isomorphic. Thus, we can refer to there being a single colimit (denoted as $\varinjlim(F)$). Also, we call this limit an injective/direct limit.

Suppose **I** is a small category and let $F : \mathbf{I} \to A$-**mod** be a functor (where A is some unital commutative ring). I'll also denote $M_I := F(I)$ for all $I \in \mathrm{Obj}(\mathbf{I})$.

Firstly, here's how to construct $\varprojlim(F)$.

Let $L \subseteq \prod_{I \in \mathrm{Obj}(\mathbf{I})} M_I$ be given by the set:

$$L = \{(x_I)_{I \in \mathrm{Obj}(\mathbf{I})} : \forall I, J \in \mathrm{Obj}(\mathbf{I}) \text{ and } \forall f \in \mathrm{Hom}_{\mathbf{I}}(I, J),\ \ (F(f))(x_I) = x_J\}.$$

I claim that L is a submodule of $\prod_{I \in \mathrm{Obj}(\mathbf{I})} M_I$. After all, $(0_{M_I})_{I \in \mathrm{Obj}(\mathbf{I})} \in N$ since all module homomorphisms send zero elements to zero elements. Also suppose $(x_I)_{I \in \mathrm{Obj}(\mathbf{I})}$, $(y_I)_{I \in \mathrm{Obj}(\mathbf{I})} \in N$ and $a, a' \in A$. Then:

$$(F(f))(ax_I + a'y_I) = a(F(f))(x_I) + a'(F(f))(y_I) = ax_J + a'y_J$$

for all $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Hence $a(x_I)_{I \in \mathrm{Obj}(\mathbf{I})} + a'(y_I)_{I \in \mathrm{Obj}(\mathbf{I})} \in N$.

Next, for every $I \in \mathbf{I}$ let π_I by the projection from $\prod_{I' \in \mathrm{Obj}(\mathbf{I})} M_{I'}$ to M_I. I shall prove that L along with all the π_I (with their domains restricted to L) forms the limit of F.

1. Suppose $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Then the following diagram holds:

$$
\begin{tikzcd}
 & & (x_I)_{I\in\mathrm{Obj}(\mathbf{I})} \arrow[dll, mapsto] \arrow[ddrr, mapsto, bend left] & & \\
 & & L \arrow[dl, "\pi_I"'] \arrow[dr, "\pi_J"] & & \\
x_I \arrow[drr, mapsto, bend right] & M_I \arrow[rr, "F(f)"] & & M_J & x_J \\
 & & & (F(f))(x_I) \arrow[r, equal] & x_J
\end{tikzcd}
$$

2. Now suppose another A-module M as well as the homomorphisms $\eta_I : M \to M_I$ satisfy the diagram that L and the π_I satisfy. By the universal property of products of modules, we know there is a unique module homomorphism $g : M \to \prod_{I\in\mathrm{Obj}(\mathbf{I})} M_I$ satisfying for all $I \in \mathrm{Obj}(\mathbf{I})$ that:

$$
\begin{tikzcd}
M \arrow[rr, bend left, "\eta_I"] \arrow[dr, dashed, "g"'] & & M_I \\
 & \prod_{I'\in\mathrm{Obj}(\mathbf{I})} M_{I'} \arrow[ur, "\pi_I"'] &
\end{tikzcd}
$$

Since our desired homomorphism must satisfy the above commuting diagram, we can conclude that our desired morphism is unique. Also note that since M and the η_I satisfy the same commuting diagram that L and the π_I satisfy, we have that $\mathrm{im}(g) \subseteq L$. So, g is our desired morphism.

Secondly, here's how to construct $\varinjlim(F)$.

For all $I \in \mathrm{Obj}(\mathbf{I})$ let $j_I : M_I \to \bigoplus_{I'\in\mathrm{Obj}(\mathbf{I})} M_{I'}$ be the embedding homomorphism. Then, define $N \subseteq \bigoplus_{I'\in\mathrm{Obj}(\mathbf{I})} M_{I'}$ to be the submodule generated by all elements of the form $j_I(x_I) - j_J((F(f))(x_I))$ where $I, J \in \mathrm{Obj}(\mathbf{I})$, $x_I \in M_I$, and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$.

We let $C := (\bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I)/N$. After letting $\pi : \bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I \to (\bigoplus_{I\in\mathrm{Obj}(\mathbf{I})} M_I)/N$ be the quotient map, we then define the map $\widetilde{j}_I := \pi \circ j_I$ for all $I \in \mathrm{Obj}(\mathbf{I})$ to be. I claim that C along with the $\widetilde{j}_I$ is the colimit of F.

1. Suppose $I, J \in \mathrm{Obj}(\mathbf{I})$ and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. Then we need to verify that the following diagram holds for all $X_I \in M_I$:

$$
\begin{tikzcd}
x_I \arrow[rrrr, mapsto] \arrow[ddd, mapsto] & & & & (F(f))(x_I) \arrow[ddd, mapsto] \\
 & M_I \arrow[rr, "F(f)"] \arrow[dr, "\widetilde{j}_I"'] & & M_J \arrow[dl, "\widetilde{j}_J"] & \\
 & & C & & \\
j_I(x_I)+N \arrow[rrrr, equal, "?"] & & & & j_J((F(f))(x_I))+N
\end{tikzcd}
$$

Fortunately, note that $j_I(x_I) - j_J((F(f))(x_I)) \in N$ because we defined N to have that element as a generator. So, the equality we needed does hold.

2. Suppose another A-module M and a bunch of homomorphisms $\eta_I : M_I \to M$ also satisfy the same commuting diagram that C and the $\widetilde{j}_I$ satisfy. Then by the universal property of direct sums of modules, there is a unique module homomorphism $h : \bigoplus_{I \in \mathrm{Obj}(\mathbf{I})} M_I \to M$ satisfying that for all $I \in \mathrm{Obj}(\mathbf{I})$ the below diagram holds:

$$\begin{tikzcd} M_I \arrow[rr, "\eta_I"] \arrow[dr, "j_I"'] & & P \\ & \bigoplus_{I' \in \mathrm{Obj}(\mathbf{I})} M_{I'} \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Yet note that $\eta_I(x_I) - \eta_J((F(f))(x_I)) = 0_P$ for all $I, J \in \mathrm{Obj}(\mathbf{I})$, $x_I \in M_I$, and $f \in \mathrm{Hom}_{\mathbf{I}}(I, J)$. As a result, $N \subseteq \ker(h)$ and in turn we can define a module homomorphism $g : C \to P$ by $g((x_I)_{I \in \mathrm{Obj}(\mathbf{I})} + N) = h((x_I)_{I \in \mathrm{Obj}(\mathbf{I})})$. This will still be the unique homomorphism satisfying that:

$$\begin{tikzcd} M_I \arrow[rr, "\eta_I"] \arrow[dr, "\widetilde{j}_I"'] & & P \\ & C \arrow[ur, dashed, "g"'] & \end{tikzcd}$$

Since A-**mod** has a limit and colimit respectively for all functors F going into A-**mod** from a small category, we say A-**mod** is complete and cocomplete respectively. Since A-**mod** is *both* complete *and* cocomplete, we also call A-**mod** a bicomplete category.

Something else I want to mention is that both of the prior constructions still work if we replace the category A-**mod** with the category A-**alg** of A-algebras and algebra homomorphisms. In the colimit construction we just have to make N the algebra generated by some points instead of the module generated by some points.

Furthermore, since **Ab** is identical to category of $\mathbb{Z}$-modules and **CRing** is identical to the category of $\mathbb{Z}$-algebras, we also now have constructions for limits and colimits of abelian groups and commutative rings.

One more colimit I want to construct is as follows. Suppose $(I, \lesssim)$ is a directed set (see my math 240b notes to recall what that means). We can turn I into a small category $\mathbf{I}$ whose objects are just the elements of I and whose morphisms are just the relations $i \lesssim j$. Now suppose $F : \mathbf{I} \to \mathbf{Set}$ is a functor. Then we can construct $\varinjlim(F)$ as follows.

Let $S_i := F(i)$ for all $i \in I$. Also, for reasons that will become apparent later, denote $F(i \lesssim j)$ as r^i_j. To construct the colimit of F, first let $S := \bigcup_{i \in I}(\{i\} \times S_i)$. Next, we use the fact that I is a directed set to define an equivalence relation. Namely, we say that $(i, x) \sim (j, y)$ if after having let $k \in I$ be any element with $i \lesssim k$ and $j \lesssim k$, we then have that there exists $z \in S_k$ with $r^i_k(x) = z = r^j_k(y)$.

I claim that $\sim$ is an equivalence relation. After all, since $i \lesssim i$ and $r_i^i = \mathrm{Id}_{S_i}$ for all $i \in I$, we know that $(i,x) \sim (i,x)$ for all $i \in I$ and $x \in S_i$. Also, its obvious from our definition that $(i,x) \sim (j,y)$ iff $(j,y) \sim (i,x)$. Finally, suppose $(i_1,x_1) \sim (i_2,x_2) \sim (i_3,x_3)$. Then we know there exists (j,y) such that $r_j^{i_1}(x_1) = y = r_j^{i_2}(x_2)$. Similarly, we know there exists (j',y') such that $r_{j'}^{i_1}(x_2) = y' = r_{j'}^{i_2}(x_3)$. And finally, we let $k \in I$ be any element with $j \lesssim k$ and $j' \lesssim k$. Note that $r_k^j \circ r_j^{i_2} = r_k^{i_2} = r_k^{j'} \circ r_{j'}^{i_2}$, and also that $r_k^{i_1} = r_k^j \circ r_j^{i_1}$ and $r_k^{i_3} = r_k^{j'} \circ r_{j'}^{i_3}$. Thus:

$$r_k^{i_1}(x_1) = r_k^j \circ r_j^{i_1}(x_1) = r_k^j \circ r_j^{i_2}(x_2) = r_k^{j'} \circ r_{j'}^{i_2}(x_2) = r_k^{j'} \circ r_{j'}^{i_3}(x_3) = r_k^{i_3}(x_3)$$

And this proves that $(i_1,x_1) \sim (i_3,x_3)$.

We now let $C := S/_{\sim}$. Also for every $i \in I$ we define $\iota_i : S_i \to C$ by $x \mapsto [(i,x_i)]$. I claim that C along with the ι_i forms a colimit.

- Note that $(i,x) \sim (j, r_j^i(x))$ for all $j \in I$ with $i \lesssim j$. Hence, the following diagram holds for all $i,j \in I$ with $i \lesssim j$:

$$\begin{tikzcd}
x \arrow[rrrr, mapsto] \arrow[ddr, mapsto] & & & & r_j^i(x) \arrow[ddl, mapsto] \\
& S_i \arrow[rr, "r_j^i"] \arrow[dr, "\iota_i"'] & & S_j \arrow[dl, "\iota_j"] & \\
& {[x]} \arrow[r, phantom, "C"] \arrow[rr, equal] & & {[r_j^i(x)]} &
\end{tikzcd}$$

- Next, suppose a set M and functions η_i also satisfy for all $i \lesssim j$ that:

$$\begin{tikzcd}
S_i \arrow[rr, "r_j^i"] \arrow[dr, "\eta_i"'] & & S_j \arrow[dl, "\eta_j"] \\
& M &
\end{tikzcd}$$

  Then by the universal property of disjoint unions of sets (i.e. that they are categorical coproducts of sets), we know that there is a unique function $h : S \to M$ such that the following diagram holds for all $i \in I$:

$$\begin{tikzcd}
& S_i \arrow[rr, "\eta_i"] \arrow[dr] & & M \\
x \arrow[dr, mapsto] & & S \arrow[ur, dashed, "\exists! h"'] & \\
& (i,x) & &
\end{tikzcd}$$

  In other words, $h(i,x) = \eta_i(x)$ for all $(i,x) \in S$. Yet now suppose $(i,x) \sim (j,y)$. Thus, there exists $k \in I$ with $i \lesssim k$, $j \lesssim k$, and $r_k^i(x) = r_k^j(y)$. Since we also know that $\eta_k \circ r_k^i = \eta_i$ and $\eta_k \circ r_k^j = \eta_j$, we can say that:

$$h(i,x) = \eta_i(x) = \eta_k \circ r_k^i(x) = \eta_k \circ r_k^j(y) = \eta_j(y) = h(j,y)$$

It follows that we can define a function $g : C \to M$ by $g([(i,x)]) := h(i,x)$. And since h was uniquely determined, we also know g is uniquely determined. This finishes showing that C is a colimit.

7/3/2026

Today I want to try and take some notes on presheaves, stalks, and germs. I will be loosely following *Differential Analysis on Complex Manifolds* by Raymond Wells. (See *citation 9*...)

To start off, suppose **C** is a category. Then a presheaf is just a functor $F : \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$. If we replace the category **Set** with a different category **D** (say A-**mod**), we just say our presheaf is **D**-valued.

A morphism between presheaves F_1 and F_2 is just a natural transformation between the two functors. In the special case that the natural transformation $\eta : F_1 \to F_2$ consists of only injective functions, then we say F_1 is a subpresheaf of F_2.

Given a topological X, a relevant category for us to define is the category $\mathbf{Op}(X)$ of nonempty open subsets of X. The objects of this category are open subsets, and the morphisms are given by set inclusions. Clearly this is a small category. Using this category, we now define a presheaf $\mathcal{F}$ *over a topological space* X to be:

- An assignment to every nonempty open set $U \subseteq X$ some other set $\mathcal{F}(U)$,
- A collection of homomorphisms $r_V^U : \mathcal{F}(U) \to \mathcal{F}(V)$ (called restriction homomorphisms) for each pair of nonempty open sets $V \subseteq U \subseteq X$ such that:
  - $r_U^U = \mathrm{Id}_{\mathcal{F}(U)}$,
  - If $W \subseteq V \subseteq U$ then $r_W^U = r_W^V \circ r_V^U$.

  As a side note, we call the objects $\mathcal{F}(U)$ sections of the presheaf $\mathcal{F}$.

The subpresheafs in this case that we care about are when we restrict the domain of $\mathcal{F}$ to only open sets containing some arbitrary point $x \in X$.

Let **D** denote either **Set**, **Ab**, **CRing**, or A-**mod** or A-**alg** (where A is some unital commutative ring). Then let $\mathcal{F}$ be a **D**-valued presheaf over a topological space X. Given any $x \in X$ we define the stalk of $\mathcal{F}$ at x to be the direct limit (i.e. colimit) of $\mathcal{F}$ restricted to the open sets containing x:

$$\mathcal{F}_x := \varinjlim_{x \in U \in \mathrm{Obj}(\mathbf{Op}(X))} \mathcal{F}(U)$$

Then, an element of $\mathcal{F}_x$ is called a germ.

The intuition for this naming scheme is that in agriculture, a sheaf refers to a bundle of wheat, then each stalk is a single wheat strand, and finally the germs are the seeds at end of the stalk. Similarly in the math sense, a sheaf (which will be a presheaf equipped with a few more assumptions) is a structure encoding local data on an entire topological space, each stalk is a structure encoding local data at a specific point, and the germs are all the possible configurations of local data at a given point.

Here are three examples of presheaves over a topological space X:

- Suppose Y is another topological space and then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ define:
$$\mathcal{F}(U) = C(U, Y) \coloneqq \{(f : U \to Y) : f \text{ is continuous}\}.$$
Also if $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is easily checked to be a presheaf which we denote $\mathcal{C}_{X,Y}$. Also if $Y = k$ (where $k = \mathbb{R}$ or $\mathbb{C}$) then this presheaf is $(k\text{-}\mathbf{alg})$-valued.

- Suppose that X in particular is a topological manifold equipped with a C^r structure (where $r \in \mathbb{N} \cup \{\infty\}$). Then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ let $\mathcal{C}_X^r(U)$ denote the space of real-valued C^r functions from U to $\mathbb{R}$. Also for $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is also easily checked to be a $(\mathbb{R}\text{-}\mathbf{alg})$-valued presheaf.
- Meanwhile, suppose that X is a topological manifold equipped with a complex structure. Then for every $U \in \mathrm{Obj}(\mathbf{Op}(X))$ let $\mathcal{O}_X$ denote the space of holomorphic functions from U. Also for $V \subseteq U \subseteq X$ let $r_V^U(f) \coloneqq f|_V$. This is also easily checked to be a $(\mathbb{C}\text{-}\mathbf{alg})$-valued presheaf.

Now a presheaf $\mathcal{F}$ over a topological space X is called a sheaf if for every collection of open sets $\{U_\alpha\}_{\alpha \in A}$ with $U \coloneqq \bigcup_{\alpha \in A} U_\alpha$ the following two axioms are satisfied:

1. If $s, t \in \mathcal{F}(U)$ satisfy that $r_{U_\alpha}^U(s) = r_{U_\alpha}^U(t)$ for all $\alpha \in A$ then $s = t$.
2. Suppose there exists $s_\alpha \in \mathcal{F}(U_\alpha)$ for all $\alpha \in A$ satisfying that if $\beta \in A$ and $U_\alpha \cap U_\beta \neq \emptyset$ then $r_{U_\alpha \cap U_\beta}^{U_\alpha}(s_\alpha) = r_{U_\alpha \cap U_\beta}^{U_\beta}(s_\beta)$. Then there exists an $s \in \mathcal{F}(U)$ such that $r_{U_\alpha}^U(s) = s_\alpha$ for all $\alpha \in A$.

Hopefully it is clear why all three examples of presheaves I just gave are also sheaves. The first bullet point is satisfied because because functions are uniquely determined by their values at each point. Meanwhile, the second bullet point is satisfied because we can glue compatible continuous, smooth, or holomorphic functions together.

7/6/2026

Augh I wanted to read the book by Wells but I've been having immense difficulty. I think my two gaps in knowledge are that I still don't have enough knowledge about manifolds, and also that I don't know any complex analysis on multivariable complex functions. Since I don't have a textbook covering the latter yet, I'll just read John Lee's *Introduction to Smooth Manifolds*.

Firstly, I want to generalize the notes about tangent planes on *pages 926 and 927* to abstract C^r manifolds.

Let M be a C^r manifold with or without boundary (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$), and consider any $p \in M$. A linear map $v : C^1(M, \mathbb{R}) \to \mathbb{R}$ is a derivation at p if

$v(fg) = f(p)v(g) + g(p)v(f)$ for all $f, g \in C^r(X, \mathbb{R})$.

Also, we let T_pM be the set of all derivations at p. Importantly, we can see that T_pM is a real vector space. So, we say an element of T_pM is a tangent vector at p.

Lemma: Suppose $v \in T_pM$.

- If $f \equiv C$ where $C \in \mathbb{R}$ then $v(f) = 0$.

  Why?
  $v(f) = \frac{1}{c}v(cf) = \frac{1}{c}v(f^2) = \frac{1}{c}(2f(p)v(f)) = 2v(f)$. By subtracting $v(f)$ from both sides we then get that $0 = v(f)$.

- If $f, g \in C^r(M, \mathbb{R})$, $f(p) = 0$, and $g(p) = 0$ then $v(fg) = 0$.

  Hopefully this is obvious.

For the sake of getting intuition for why derivations should be thought of as tangent vectors, suppose we have a k-dimensional C^r manifold $M \subseteq \mathbb{R}^n$, and for the sake of convenience assume M has no boundary. (Here I'm taking M to be a manifold as was studied on *pages 77-99* as opposed to the more general abstract manifolds covered by John Lee. As a side note, M would also be an abstract manifold and its C^r structure would be generated by all full rank bijective C^r maps from open subsets of $\mathbb{R}^k$ to open subsets of M which also have continuous inverses.)

Next, consider any $p \in M$ and let ϕ be a C^r diffeomorphism from an open set U of $\mathbb{R}^k$ to an open set V' of M containing p. Also let $q = \phi^{-1}(p)$. Then recalling *pages 926 and 927*, we define the "geometric tangent plane" based at p as:

$$\widetilde{T_pM} := \{(p, \vec{y}) : \vec{y} \in \mathrm{im}(\mathrm{D}\phi(q) : \mathbb{R}^k \to \mathbb{R}^n)\}$$

Importantly, $\widetilde{T_pM}$ is an k-dimensional vector space. Also suppose that $\vec{v}_p = (p, \vec{v}) \in \widetilde{T_pM}$.

I realize I have two diffent definitions for what it means for a function $f : M \to \mathbb{R}$ to be a C^s function (where $s \leq r$). So I want to quickly rectify that.

- Definition 1: $f : M \to \mathbb{R}$ is a C^s map if there is an open set $W \subseteq \mathbb{R}^n$ containing M and a C^s function $g : W \to \mathbb{R}$ with $g|_M = f$.
- Definition 2: f is a C^s function in the manifold sense. Equivalently, this means for any open sets $U \subseteq \mathbb{R}^k$ and $V \subseteq M$ as well as any bijective C^r function $\varphi : U \to V$ such that $\mathrm{D}\varphi(x)$ has full rank for all $x \in U$ and φ^{-1} is continuous, we have that $f \circ \varphi$ is a C^s map. (Note: this is equivalent to the manifold definition because the φ form a C^r subatlas of the C^r structure of M).

  $(1 \implies 2)$
  This is hopefully obvious.

  $(2 \implies 1)$
  Recall from lemma 23.1 on *page 83* that by passing to a partition of unity, it suffices to show definition 1 only locally. In other words, we must show for any $p \in M$ that there exists a C^s function $g : W \to \mathbb{R}$ for some open set $W \subseteq \mathbb{R}^n$ containing p such that $g|_{M\cap V} = f|_{M\cap V}$.

  Fortunately, we can find open sets $U \subseteq \mathbb{R}^k$ and $V \subseteq M$ as well as a homeomorphism $\varphi : U \to V$ such that $p \in V$, φ is a C^r function, and $\mathrm{D}\varphi(x)$ has full rank for all $x \in U$. Next, note from theorem 24.1 on *page 87* that φ^{-1} can be extended to a C^r function $\psi : W \to \mathbb{R}^k$ such that $W \subseteq \mathbb{R}^n$ is open and $\psi|_V = \varphi^{-1}$. But now as ψ is continuous and $\psi(V) = \varphi^{-1}(V) = U$, we know that $\psi^{-1}(U)$ is an open subset of W containing V. Plus, $p \in V \subseteq \psi^{-1}(U)$. In other words, we can without loss of generality shrink W so that $\psi(W) \subseteq U$ and still have that $p \in W$. And finally, $(f \circ \varphi) \circ \psi = f$ on M and $(f \circ \varphi) \circ \psi$ is a composition of a C^s map and a C^r map on W. Hence, we've proven that definition 1 holds locally for if f.

Next, suppose $f \in C^1(M, \mathbb{R})$. We define its directional derivative $\mathrm{D}_{\vec{v}}|_p f$ in the direction of $\vec{v}$ as follows:

Let $\gamma : [-1, 1] \to M$ be any C^1-path such that $\gamma(0) = p$ and $\gamma'(0) = \vec{v}$.
As for how we know such a path γ exists, recall that $\vec{v} \in \mathrm{D}\phi(q)$. Hence, there exists $w \in \mathbb{R}^k$ such that $(\mathrm{D}\phi(q))\vec{w} = \vec{v}$. Let $\widetilde{\gamma} : [-1, 1] \to U$ be a C^1-path such that $\widetilde{\gamma}(0) = q$ and $\widetilde{\gamma}'(0) = \vec{w}$. Then let $\gamma := \phi \circ \widetilde{\gamma}$. Clearly $\gamma(0) = p$ and $\gamma([-1, 1])$ is contained in M. Also by chain rule we have that:

$$\gamma'(0) = (\mathrm{D}\phi(q))\widetilde{\gamma}'(0) = (\mathrm{D}\phi(q))\vec{w} = \vec{v}.$$

Now suppose $f \in C^r(M)$. we define $\mathrm{D}_{\vec{v}}|_p f = (f \circ \gamma)'(0)$. I claim this definition is independent of the specific path γ chosen. After all:
Suppose γ_1 and γ_2 are two distinct C^1-paths such that $\gamma_i(0) = p$ and $\gamma_i'(0) = \vec{v}$ for both i. Then after extending f to a C^r function on an open set $V \subseteq \mathbb{R}^n$ with $V' = V \cap M$ (which we can do since $f \in C^r(M, \mathbb{R})$), we know that:

$$(f \circ \gamma_1)'(0) = \nabla f(p) \cdot \gamma_1'(0) = \nabla f(p) \cdot \vec{v} = \nabla f(p) \cdot \gamma_2'(0) = (f \circ \gamma_2)'(0).$$

It's also easy to see that the directional derivative $\mathrm{D}_{\vec{v}}|_p$ is a derivation at p. After all, we can just apply the product rule to the gradiant operator and also note that taking a dot product against a fixed vector $\vec{v}$ defines a linear functional.

Now consider the function $\widetilde{T_pM}$ to T_pM given by $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$. We can see from the fact that taking a dot product against a fixed vector $\nabla f(p)$ defines a linear functional that our mapping $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ is actually a linear map between vector spaces.

Furthermore, I claim that our linear map is injective. After all, suppose $(p, \vec{v})$ maps to the zero derivation (i.e. the derivation mapping every C^r function to 0). Then let $(p, u_1), \ldots, (p, u_k)$ be an orthonormal $\mathbb{R}$-basis for the vector space $\widetilde{T_pM}$. We can uniquely express $\vec{v} = c_1 u_1 + \ldots + c_k u_k$ where $c_1, \ldots, c_k \in \mathbb{R}$. And now for any $f \in C^1(M, \mathbb{R})$ we have that:

$$\mathrm{D}_{\vec{v}}|_p f = \sum_{j=1}^{k} c_j \mathrm{D}_{u_j}|_p f$$

To finish off, we can define $f_i \in C^1(M, \mathbb{R})$ by $f_i(x) = x \cdot u_i$ for all $x \in M \subseteq \mathbb{R}^n$. This is a C^1 map because it more strongly extends to a smooth map on all of $\mathbb{R}^n$. Furthermore, note that $\nabla f_i(p) = u_i$. So, $\mathrm{D}_{u_j}|_p f = u_i \cdot u_j$. And this equals 1 if $i = j$ and 0 otherwise. So, we can conclude for all i that:

$$0 = \mathrm{D}_{\vec{v}}|_p f_i = \sum_{j=1}^{k} c_j \mathrm{D}_{u_j}|_p f_i = c_i$$

In other words $\vec{v} = 0u_1 + \ldots + 0u_k = 0$. And with that we've proven that $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ has a trivial kernel.

Now to finish off, note that $\widetilde{T_pM}$ is k-dimensional. We'll also soon show for a general abstract C^r k-dimensional manifold that T_pM is k-dimensional. Hence, our injective linear map must also be surjective. And thus, we've proven that our linear map $\widetilde{T_pM}$ to T_pM given by $(p, \vec{v}) \mapsto \mathrm{D}_{\vec{v}}|_p$ is a bijection.

The T.L.D.R. to all of this is that you should think of a tangent plane as being isomorphic to a vector space of directional derivative operators. Also, derivations are a generalization of directional derivatives.

Suppose M and N are C^r manifolds with or without boundary, and $F : M \to N$ is a C^r map. We define a map $\mathrm{d}F_p : T_pM \to F_{F(p)}N$ called the differential of F at p such that $\mathrm{d}F_p(v)f \coloneqq v(f \circ F)$ for all $v \in T_pM$ and $f \in C^1(N, \mathbb{R})$.

> To see that this map is well-defined, suppose $v \in T_pM$ and let $f, g \in C^r(N, \mathbb{R})$. Also suppose $C \in \mathbb{R}$. Then $\mathrm{d}F_p(v)(cf) = v(cf \circ F) = cv(f \circ F) = c\mathrm{d}F_p(v)f$. Similarly,
>
> $$\begin{aligned} \mathrm{d}F_p(v)(f+g) &= v((f+g) \circ F) \\ &= v((f \circ F) + (g \circ F)) = v((f \circ F)) + v((g \circ F)) \\ &= \mathrm{d}F_p(v)f + \mathrm{d}F_p(v)g \end{aligned}$$
>
> Thus, we've shown that $\mathrm{d}F_p(v)$ is a linear map on $C^1(N, \mathbb{R})$. As for additionally showing that $\mathrm{d}F_p(v)$ is a derivation at $F(p)$, note that:
>
> $$\begin{aligned} \mathrm{d}F_p(v)(fg) &= v((fg) \circ F) \\ &= v((f \circ F)(g \circ F)) = (f \circ F)(p)v(g \circ F) + (g \circ F)(p)v(f \circ F) \\ &= f(F(p))\mathrm{d}F_p(v)g + g(F(p))\mathrm{d}F_p(v)f \end{aligned}$$

Proposition 3.6: Let M, N, and P be C^r manifolds with or without boundary, let $F : M \to N$ and $G : N \to P$ be C^r maps, and let $p \in M$

(a) $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is a linear map.

Proof:
Suppose $a, b \in \mathbb{R}$ and $v, w \in T_pM$. Then for any $f \in C^1(N)$ we have that:

$$\begin{aligned}\mathrm{d}F_p(av + bw)f &= (av + bw)(f \circ F)\\ &= av(f \circ F) + bw(f \circ F) = a\mathrm{d}F_p(v)f + b\mathrm{d}F_p(w)f\end{aligned}$$

(b) $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$.

Proof:
Let $v \in T_pM$ and $f \in C^1(P, \mathbb{R})$. Then:

$$\mathrm{d}(G \circ F)_p(v)f = v(f \circ G \circ F) = (\mathrm{d}F_p(v))(f \circ G) = (\mathrm{d}G_{F(p)}(\mathrm{d}F_p(v)))f.$$

Thus $\mathrm{d}(G \circ F)_p(v) = \mathrm{d}G_{F(p)}(\mathrm{d}F_p(v))$.

(c) $\mathrm{d}(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM}$.

Proof:
Let $v \in T_pM$ and $f \in C^1(M, \mathbb{R})$. Then:

$$(\mathrm{d}(\mathrm{Id}_M)_p v)f = v(f \circ \mathrm{Id}_M) = v(f).$$

As a result, $\mathrm{d}(\mathrm{Id}_M)_p v = v$. In other words, $\mathrm{d}(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM}$.

(d) If F is a C^r-diffeomorphism from M to N (i.e. a bijection such that F and F^{-1} are C^r maps) then $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is an isomorphism and $(\mathrm{d}F_p)^{-1} = \mathrm{d}(F^{-1})_{F(p)}$.

Why?
From parts (b) and (c) we know that:

- $\mathrm{Id}_{T_pM} = \mathrm{d}(\mathrm{Id}_M)_p = \mathrm{d}(F^{-1} \circ F)_p = \mathrm{d}(F^{-1})_{F(P)} \circ \mathrm{d}F_p$,
- $\mathrm{Id}_{T_{F(p)}N} = \mathrm{d}(\mathrm{Id}_N)_{F(p)} = \mathrm{d}(F \circ F^{-1})_{F(p)} = \mathrm{d}F_p \circ \mathrm{d}(F^{-1})_{F(p)}$. ■

To make use of the differential of a map $F : M \to N$ between C^r manifolds at a point p, we need to be able consider functions defined on only neighborhoods of $F(p)$ instead of all N. There are two common approaches to this conundrum. One approach is to redefine derivations in terms of germs at a point p. This is the only approach available for complex manifolds (because the identity principle thwarts us). The other approach is to bring in "bump" functions. For the time being, I'm going to follow the other approach. That said, both approaches will eventually be shown to give isomorphic constructions.